3.1.1 Atomic structure: Sheet 5

Electron configuration and ionisation energy · 16 marks · about 25 minutes

Key information

- Ionisation energy: energy needed to remove one mole of electrons from one mole of gaseous atoms (or ions).
- Successive IE equations always show gaseous ions losing one electron, eg $Na^+(g) \rightarrow Na^{2+}(g) + e^-$
- Fill sub-levels 1s 2s 2p 3s 3p 4s 3d 4p. Cr and Cu are exceptions ($3d^5 4s^1$, $3d^{10} 4s^1$); for transition metal ions, remove 4s electrons first.
- Across a period: IE increases (more protons, same shielding). Down a group: IE decreases (outer electron further away, more shielding).
- Dips: Group 3 (electron in higher-energy p sub-level) and Group 6 (paired p electrons repel).
- A big jump in successive IEs means the electron is now coming from a shell closer to the nucleus.

Section A: Self-quiz

01 Complete the full electron configuration of a chromium atom.

$1s^2$

[1 mark]

02 Give the full electron configuration for Br

[1 mark]

03 What is the electron configuration of V^{2+} in the ground state?

[1 mark]

A $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^3$

B $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^1\ 4s^2$

C $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^3\ 4s^2$

D $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^5\ 4s^2$

04 Which equation shows the process that occurs during the second ionisation of magnesium?

[1 mark]

A $Mg(g) \rightarrow Mg^+(g) + e^-$

B $Mg(g) \rightarrow Mg^{2+}(g) + 2e^-$

C $Mg^+(g) \rightarrow Mg^{2+}(g) + e^-$

D $Mg^{2+}(g) \rightarrow Mg^{3+}(g) + e^-$

05 Which element has a first ionisation energy lower than that of sulfur?

[1 mark]

A Chlorine

B Oxygen

C Phosphorus

D Selenium

Section B: Exam questions

06 The first ionisation energies of the elements in Period 2 change as the atomic number increases.

Explain the pattern in the first ionisation energies of the elements from lithium to neon.

[6 marks]

07 This question is about atomic structure.

07.1 There is a general trend for an increase in ionisation energy across Period 3.

Give one example of an element that deviates from this trend.

Explain why this deviation occurs.

[3 marks]

07.2 Give an equation, including state symbols, to represent the process that occurs when the third ionisation energy of sodium is measured.

[1 mark]

08 Identify the Group 2 metal (Mg to Ba) with the lowest second ionisation energy.

[1 mark]

Mark Scheme

M1, M2… are separate marking points. / separates acceptable alternatives. Words in brackets are not needed for the mark. ECF = error carried forward. Questions and mark schemes © AQA, reproduced for internal school use.

Question	Marking guidance	Comments	Mark
01	$(1s^2)\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^5\ 4s^1$	Source: 7404/1 Jun 2019 Q03.1 Or $(1s^2)\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^1\ 3d^5$ Ignore commas; do not penalise capitals and subscripts	1
02	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}\ 4s^2\ 4p^5$	Source: 7404/1 Jun 2024 Q06.1 Allow $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^{10}\ 4p^5$	1
03	**A**	Source: 7404/1 Nov 2021 Q12	1
04	**C**	Source: 7404/1 Jun 2024 Q22	1
05	**D**	Source: 7404/1 Nov 2021 Q24 Selenium	1
06	**Level 3 (5–6 marks):** All stages covered with matching justifications; explanation of each stage generally correct and virtually complete. Well structured, no repetition or irrelevant points; accurate use of technical terms. **Level 2 (3–4 marks):** Two stages with matching justifications OR three stages with incomplete justifications. Some attempt at structure; reasonable clarity; some minor errors in technical terms. **Level 1 (1–2 marks):** One stage with matching justification OR two stages with incomplete justifications. Isolated statements, not in a logical order or confused reasoning. **Level 0:** Insufficient correct chemistry to gain a mark.	Source: 7404/1 Jun 2019 Q04 **Stage 1: General trend (Li → Ne)** 1a 1st IE increases; 1b more protons / increased nuclear charge; 1c electrons in same energy level / shell; 1d no extra / similar shielding; 1e stronger attraction between nucleus and outer e^- OR outer e^- closer to nucleus (ignore radius decreases) **Stage 2: Deviation Be → B** 2a B lower than Be; 2b outer electron in (2)p; 2c higher in energy than (2)s. If Al vs Mg then do not award 2a or 2b **Stage 3: Deviation N → O** 3a O lower than N; 3b 2 electrons in (2)p need to pair; 3c pairing causes repulsion (not if clearly in s orbital). If S vs P then do not award 3a or 3b	6
07.1	M1 Aluminium / Al M2 (Outer) electron in (3)p orbital / sub-shell (level) M3 (3p) higher in energy / slightly more shielded (than 3s) / slightly further away (than 3s) OR M1 Sulfur / S M2 (Outer) electrons in (3)p orbital begin to pair M3 Repel	Source: 7404/1 Nov 2020 Q01.1–01.2 Allow M2/M3 if a Group 3 (or Group 6) element is given M2 not energy level Do not allow just p^4 vs p^3	3
07.2	$Na^{2+}(g) \rightarrow Na^{3+}(g) + e^-$	State symbols essential Allow $Na^{2+}(g) + e^- \rightarrow Na^{3+}(g) + 2e^-$	1
08	Ba (barium)	Source: 7404/1 Jun 2025 Q04.5 Allow Ra (radium)	1